

HERA

Humanoid Engineering & Response Automaton

Industrial Warehouse Humanoid Robot — Design Concept Proposal

Mini Project Report—Task 4

CodeAlpha Robotics & Automation Internship

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ABSTRACT

This report presents HERA, Humanoid Engineering & Response Automaton, is a design concept for a bipedal humanoid robot built for industrial warehouse environments. The proposal covers mechanical structure, joint design, actuator selection, sensor systems, and control architecture. HERA targets 34 degrees of freedom, 25 kg payload capacity, and 4-hour battery life, benchmarked against current platforms including Tesla Optimus Gen 2 and Boston Dynamics Atlas.

1. Introduction

Most warehouse automation today uses either fixed robotic arms or wheeled mobile robots. Both have real limits: fixed arms need purpose-built workstations, and wheeled robots only interact with the floor plane. Neither can climb a ladder, use standard shelving, or operate tools designed for human hands. A humanoid robot built to human proportions can work in any environment a human can without requiring infrastructure changes. That is the core motivation for HERA.

The global warehouse automation market exceeded \$23 billion in 2025. Labor shortages in physically demanding warehouse roles are accelerating demand for flexible robotic solutions. HERA is designed to fill that gap: a general-purpose physical agent that handles heavy loads, navigates tight aisles, and works safely alongside people.

2. Concept Overview & Specifications

HERA stands 1.75 m tall and weighs 62 kg sized to fit standard warehouse aisle widths, shelf heights, and conveyor dimensions without any facility modifications. The design has three core priorities: safety near humans, high payload capability, and zero infrastructure requirements.

Spec	Value	Notes
Height / Weight	1.75 m 62 kg	Human-scale warehouse fit
Total DOF	34	Full body articulation
Payload per arm	25 kg	Heavy warehouse loads
Walking speed	1.5 m/s cruise	2.0 m/s max
Battery life	~4 hours	Hot-swappable Li-ion pack
Compute	NVIDIA Jetson Orin + FPGA	High-level AI + 1 kHz real-time control

Table 1: HERA top-level specifications

3. Mechanical Design

3.1 Structure & Materials

The primary skeleton uses carbon fiber reinforced polymer (CFRP) 5× stronger than steel at one third the weight. Joint housings are titanium alloy (Ti-6Al-4V) for fatigue resistance under cyclic loading. Outer panels are 6061-T6 aluminum for cost-effective lightweight coverage. This material hierarchy keeps total robot mass at 62 kg while maintaining the structural rigidity needed for 25 kg payload handling.

3.2 Degrees of Freedom

HERA has 34 total degrees of freedom distributed across the full body. The DOF count is higher than most current platforms specifically to enable dexterous manipulation the tendon-driven 5 DOF hands are what allow HERA to grip boxes, operate tools, and handle irregular objects that simpler grippers cannot manage.

Body Region	DOF	Actuator	Role
Head & Neck	3	DC servo	Camera pan/tilt, situational awareness
Arms (×2)	7ea	SEA	Reach, lift, manipulate — 25 kg payload
Hands (×2)	5ea	Tendon servo	Dexterous grip, tool use
Torso	3	High-torque SEA	Bend, twist, weight shifting
Legs (×2)	6ea	SEA	Walking, stair climbing, balance
TOTAL	34	—	—

Table 2: DOF breakdown by body region

3.3 Actuators

The most important mechanical decision in this design is the actuator choice. HERA uses Series Elastic Actuators (SEAs) for all major joints. A SEA places a calibrated spring between the motor and the joint output. This does three things: it measures the exact torque being applied (via spring deflection), it absorbs shock loads that would damage rigid gearboxes, and it makes the joint back-drivable meaning if a human pushes against HERA's arm, it yields rather than rigidly resisting. This is a fundamental safety requirement for a robot working next to people. Standard rigid servos are faster

and cheaper, but they cannot provide these safety guarantees, which is why SEAs are the right trade-off here.

3.4 Balance & Locomotion

Bipedal balance is controlled using the Zero Moment Point (ZMP) algorithm a standard technique in humanoid robotics that calculates where the net ground reaction force acts. As long as the ZMP stays within the robot's foot contact area, the robot is stable. The control loop runs at 1 kHz on a dedicated FPGA, continuously adjusting joint torques to maintain balance during walking, load carrying, and reaching movements. Feet are 200 × 120 mm with compliant rubber soles to handle the uneven surfaces common in warehouse floors.

4. Sensors & Control

4.1 Sensor Suite

HERA uses: a head-mounted 3D LiDAR for 360° environment mapping and obstacle detection up to 30 m; stereo RGB-D cameras for object recognition and depth; a 6-axis IMU feeding the ZMP balance controller; force/torque sensors at both wrists and ankles for safe payload handling; and 34 joint encoders for closed-loop position control. The combination gives HERA both the long-range awareness needed for navigation and the close-range sensitivity needed for safe manipulation.

4.2 Control Architecture

High-level reasoning (SLAM navigation, object detection, task planning) runs on an NVIDIA Jetson Orin NX. Real-time joint control at 1 kHz runs on a dedicated FPGA. Both communicate over Ether CAT. Software is built on ROS 2 with Nav2 for navigation and Move It 2 for motion planning. Safety systems include hardware e-stops on both forearms, speed limiting near humans (0.5 m/s within 2 m, 0.1 m/s within 1 m), force-limited compliant mode on unexpected contact, and a controlled sit-down fail-safe on critical fault.

5. Comparison with Existing Platforms

Spec	HERA	Tesla Optimus G2	BD Atlas	Agility Digit
Height	1.75 m	1.73 m	1.50 m	1.75 m
Weight	62 kg	57 kg	89 kg	65 kg

DOF	34	28	28	30
Payload	25 kg	20 kg	11 kg	16 kg
Actuator	SEA	Linear elec.	Hydraulic	SEA
Status	Concept	Pilot	Research	Commercial

Table 3: HERA vs current humanoid platforms (public specs, 2025–2026)

HERA's key differentiator is the combination of higher DOF and higher payload vs current platforms. The 34 DOF (vs 28–30 for competitors) comes from the full 5 DOF tendon hands. The 25 kg payload (vs 11–20 kg) is enabled by the SEA actuator choice, which supports higher continuous torque. HERA is a design concept these are targets, not verified figures.

6. Challenges & Conclusion

The main honest challenges are: 34 DOF means 34 failure points mechanical complexity is a real reliability risk. SEAs have lower bandwidth than rigid actuators, limiting fast-motion tasks. Battery life drops significantly under continuous heavy lifting. And a prototype would cost \$200,000–\$400,000 at current component prices.

That said, the humanoid warehouse robot space is moving fast Agility Robotics is already running commercial pilots, Tesla is expanding its Optimus program, and Aurora is deploying autonomous freight trucks. HERA represents one design point in that space: prioritizing safety and dexterity over simplicity, with the SEA actuator as the central engineering decision that ties those priorities together.

References

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